Chapter 3 . Strings.

- String is a data type in Python
- String is a sequence of characters enclosed in quotes.
- We can primarily write a string in these three ways.

1. Single quoted string → a = 'Harry'
2. Double quoted string → b = "Harry"
3. Triple quoted string → c = '''Harry'''
   ↓ used for multiline strings

String Slicing

A string in Python can be sliced for getting a part of the string.

Consider the following string:

name = "

H	a	r	r	y
0	1	2	3	4
(-5)	(-4)	(-3)	(-2)	(-1)

" ⇒ length = 5

The index in a string starts from 0 to (length - 1) in Python. In order to slice the string, we use the following syntax:

sl = name [ind_start : ind_end]

first index included   last index is not included

sl [0:3] returns "Har" → characters from 0 to 3
sl [1:3] returns "ar" → characters from 1 to 3

Negative indices: Negative indices can also be used as shown in the figure above. -1 corresponds to the (length -1) index, -2 to (length -2)

Slicing with skip value:
We can provide a skip value as a part of our slice like this:

```
word = "amazing"
word [1:6:2] → ~~mnz~~ 'mzn'
```

Other advanced slicing ~~teq~~ techniques

```
word = "amazing"
word [:7] → word [0:7] → 'amazing'
word [0:] → word [0:7] → 'amazing'
```

String functions
Some of the mostly used functions to perform operations on or manipulate strings are:

1> len() function → This function returns the length of the string.

```
len ("Harry") → return 5
```

2> string.endswith("rry") → This function tells whether the variable string ends with the string "rry" or not. if string is "Harry", it returns true for "arr" since Harry ends with rry.

3> string. Count ("i") → Counts the total number of occurence of any character.

4> String. Capitalize () → This function capitalizes the first character of a given string.

5> String. find (word) → This function finds a word and returns the index of first occurence of that word in the string.

Escape Sequence Characters.

Sequence of characters after backslash '\' → Escape sequence character.

Escape sequence character comprises of more than one characters but represents one character when used within the strings.

Examples \n , \t , \' , \\ etc.

\n → new line, \t → Tab, \' → Single Quote, \\ → backslash